Unit 1 – TQM Fundamentals: Priority-Ordered Exam Notes

19MEE347 Total Quality Management · Built from your Unit 1 slides (87 slides) and your priority table

How to use these notes

The notes follow your priority list. Study Section A completely, Section B for the key points only, and Section C by skimming. Every topic carries an importance tag:

- 🔴 **VERY IMPORTANT** – learn fully, practise writing it
- 🟡 **MODERATELY IMPORTANT** – know the key points and one example
- 🟢 **JUST READ** – skim once, don't memorise

Marks and likelihoods below are **my estimates** from your priority table, Assignment 1 and the quiz, not guaranteed exam questions. Unit 1 is expected to carry roughly 20–25 marks of the 100.

Priority map

#	Topic	Importance	Why	Suggested share of your Unit 1 time
A1	Deming's philosophy and 14 points	🔴 VERY IMPORTANT	In your table; Assignment 1 Q2; quiz Set B Q1 (repeat topic)	20%
A2	Dimensions of quality	🔴 VERY IMPORTANT	In your table; Assignment 1 Q1	12%
A3	Quality costs (4 types)	🔴 VERY IMPORTANT	In your table (concept + possible numerical); Assignment 1 Q1	15%
A4	Leadership	🔴 VERY IMPORTANT	In your table; Assignment 1 Q2	6%
A5	Quality council	🔴 VERY IMPORTANT	In your table; Assignment 1 Q2	6%

#	Topic	Importance	Why	Suggested share of your Unit 1 time
A6	Barriers to TQM implementation	🔴 VERY IMPORTANT	In your table; Assignment 1 Q2	6%
B1	Quality planning	🟡 MODERATE	In your table	5%
B2	Historical review of TQM	🟡 MODERATE	In your table	4%
B3	Principles of TQM	🟡 MODERATE	In your table	5%
B4	Quality statements (vision, mission, policy)	🟡 MODERATE	In your table; Assignment 1 Q2	5%
B5	Strategic planning	🟡 MODERATE	In your table; Assignment 1 Q2	5%
B6	Three spheres of quality	🟡 MODERATE	Classroom quiz Set A Q1	6%
C1	Definition of quality	🟢 JUST READ	In your table	3%

SECTION A – 🔴 VERY IMPORTANT

A1. Deming's philosophy and 14 points 🔴 (highest value in Unit 1)

Why it matters: it is asked in your Assignment 1 Q2 and quiz Set B Q1 ("explain Deming's philosophy and discuss any five of the 14 points"). One well-prepared answer covers both.

Who: W. Edwards Deming (1900–1993), American engineer, statistician and consultant. He learnt statistical quality control from Shewhart and taught SPC/SQC to Japanese engineers and CEOs in 1950.

His definition: quality is a predictable degree of uniformity and dependability, at low cost and suited to the market. He also said a product has quality if it helps somebody and enjoys a good, sustainable market.

Deming's chain reaction (the "philosophy" part of the answer): Improve quality → less rework and fewer mistakes, so cost falls → productivity improves → capture the market with better quality and lower price → stay in business → long-term competitive strength

and jobs.

System of Profound Knowledge (4 parts):

1. Appreciation for a system – optimise the whole system, not individual parts; it needs cooperation.
2. Psychology – decisions are made and carried out by people, so understand them.
3. Understanding variation – two types: common causes (built into the system, reduced only by changing the system/technology) and special causes (find and remove these).
4. Theory of knowledge – decisions should rest on facts, data and sound theory, not management fads.

PDSA cycle: Plan → Do → Study → Act, repeated for continual improvement (the Shewhart cycle in your slides is called Plan/Do/Check/Act).

The 14 points – one-line memory version

#	Point	Core idea (learn this line)
1	Constancy of purpose	Plan for long-term improvement and survival, not just short-term profit
2	Adopt the new philosophy	We can no longer accept delays, mistakes and defects as normal
3	Cease dependence on mass inspection	Build quality in; require statistical evidence instead of inspecting everything
4	End lowest-tender contracts	Judge suppliers on price **and** quality; move toward single suppliers with long-term trust
5	Improve every process	Constantly and forever improve the system; this reduces cost (use the seven QC tools, PDCA)
6	Institute training on the job	Modern on-the-job training for everyone, including management
7	Institute leadership	Supervisors should help people and machines do a better job, not punish
8	Drive out fear	Two-way communication so people work effectively; fear causes stress and absenteeism
9	Break down barriers	Research, design, sales and production must work as one team

#	Point	Core idea (learn this line)
10	Eliminate slogans and exhortations	Posters and zero-defect targets don't help because most problems lie in the system
11	Eliminate arbitrary numerical targets	Replace quotas and management by numbers with leadership and statistical methods
12	Permit pride in workmanship	Remove barriers (e.g. annual merit rating) that rob workers of pride
13	Encourage education	A vigorous programme of education and self-improvement for all
14	Top management commitment	The transformation is everybody's job, but top management must act every day on points 1–13

Which five to write if the question says "any five": choose **3, 4, 5, 8 and 14** (or 1, 3, 5, 8, 14). Your slides give the most supporting detail for these, so each is easy to expand to 4–5 lines.

Seven deadly diseases (add as a bonus paragraph if time allows): lack of constancy of purpose; emphasis on short-term profits; evaluation by performance/merit rating/annual review; mobility of management; running a company on visible figures alone; excessive medical costs; excessive warranty costs fuelled by contingency-fee lawyers.

Answer template (8 marks):

1. 3 lines: who Deming is and his definition of quality
2. Chain reaction (as a small flow diagram)
3. System of Profound Knowledge (4 bullets)
4. Five points × 4–5 lines each, with a one-line example
5. 1 line conclusion: the 14 points are management's responsibility, achieved through PDSA and leadership

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A2. Dimensions of quality 🔴

Why it matters: Assignment 1 Q1 asks you to analyse the role of quality dimensions in quality planning.

Product (manufacturing) dimensions – your slides list nine

Dimension	Meaning	Example from your slides (mobile phone)
Performance	Primary characteristics	Audio quality
Features	Secondary characteristics	Camera, music system, social media
Reliability	Consistency of performance; measured by MTBF	MTBF in months
Conformance	Meeting national, international and industry standards	Phones banned for failing regulations
Durability	Useful life	Low-price phone ≈ 2–3 years
Serviceability	Ease and speed of after-sales service or repair	Faster, economical service
Responsiveness	Ability to adapt to changing customer needs	Dual-SIM option
Aesthetics	Appearance, feel, smell, taste, colour	Slim model, touch screen
Reputation	Brand name, perceived quality	Apple, Samsung

Memory hook: **P F R C D S R A R** (Performance, Features, Reliability, Conformance, Durability, Serviceability, Responsiveness, Aesthetics, Reputation). The textbook list of eight usually leaves out Responsiveness, but your slides include it, so write nine if your answer follows your slides.

Service dimensions (7)

Dimension	Question it answers	Example
Time and timeliness	How long is the wait; is it on time?	Overnight package delivered overnight
Completeness	Is everything asked for provided?	Complete mail order
Courtesy	How are customers treated?	Pleasant phone operators
Consistency	Same service every time?	Newspaper on time every morning

Dimension	Question it answers	Example
Accessibility and convenience	How easy is it to get service?	Calls answered quickly
Accuracy	Is it done right every time?	Correct bank statement
Responsiveness	Reaction to unusual situations	Operator handling unusual questions

How to link dimensions to quality planning (for the Assignment 1 Q1 style question): the organisation first identifies which dimensions matter to its customers, converts them into measurable specifications, and then assigns resources for design, process control and inspection. These are the inputs that drive quality costs (A3).

A3. Quality costs 🔴 (concept + possible numerical)

Why it matters: your priority table marks it as concept plus possible numerical, and Assignment 1 Q1 explicitly asks how the four costs influence organisational performance.

Two groups:

- **Cost of achieving good quality** = prevention + appraisal
- **Cost of poor quality** = internal failure + external failure

Category	What it is	Items (from your slides)
Prevention	Spent to stop defects occurring	Quality planning, product design, process control, training, quality information systems
Appraisal	Spent to measure and check quality	Inspection and testing, test-equipment upkeep, operator time on data gathering and adjustments
Internal failure	Defects found **before** delivery	Scrap, rework, process-failure investigation, process downtime, price downgrading (“seconds”)
External failure	Defects found **after** delivery	Customer complaints, product returns, warranty claims, product liability, lost sales

Measuring and reporting – index numbers (ratios of quality cost to a base):

- Labour index = quality cost ÷ labour hours

- Cost index = quality cost ÷ manufacturing cost
- Sales index = quality cost ÷ sales
- Production index = quality cost ÷ units of final product

Total cost of quality = Prevention + Appraisal + Internal failure + External failure

Crosby's version in your slides: COQ = price of non-conformance + price of conformance. Juran's point: speak to senior management in money, i.e. the cost of poor quality.

How the four costs affect organisational performance (key argument for Assignment 1 Q1):

- More spending on prevention and appraisal usually **reduces** internal and external failure costs.
- Failure costs are the most damaging: external failure also loses customers and reputation.
- Best strategy is to move spending toward prevention so total quality cost falls while customer satisfaction, productivity and profit rise.

Mini numerical (my own practice example, not from your slides): Prevention ₹2 lakh, appraisal ₹3 lakh, internal failure ₹4 lakh, external failure ₹6 lakh; sales ₹300 lakh. Total COQ = 2 + 3 + 4 + 6 = ₹15 lakh. Sales index = 15 ÷ 300 = 0.05 = 5% of sales. Cost of poor quality = 4 + 6 = ₹10 lakh (two-thirds of the total, a sign that prevention is under-funded).

A4. Leadership 🔴

Role of TQM leaders (from your slides):

- Everyone is responsible for quality improvement, especially senior management and the CEO.
- Practise "managing by walking around" (MBWA).
- Make sure team decisions match the organisation's quality statements.
- Read TQM literature and attend conferences to keep up with tools and methods.
- Take part in award and recognition ceremonies celebrating quality successes.
- Coach others and teach in TQM seminars.
- Collaborate with internal and external customers and suppliers through visits, focus groups and surveys.
- Live and communicate TQM.

TQM implementation starts at the top: begins with senior managers and the CEO; choose timing; form the quality council; involve union leaders; train everyone in quality awareness and problem solving; the council decides quality improvement projects.

Links to remember: Deming point 7 (institute leadership) and point 14 (top management commitment); Crosby's view that quality can't be delegated.

A5. Quality council 🔴

Composition: CEO and senior managers of the functional areas (research, manufacturing, finance, sales, marketing), plus one coordinator and a union representative.

Duties:

1. Develop the quality statements: vision, mission, quality policy, core values
2. Develop strategic long-term plans and annual quality improvement programmes
3. Make a quality training programme
4. Monitor the costs of poor quality
5. Determine the performance measures of the organisation
6. Find projects that improve processes and produce customer satisfaction
7. Establish work-group teams and measure their progress
8. Establish and review the recognition and reward system

Link: the CEO writes the quality policy after studying workforce feedback, and the quality council must approve it.

Memory hook: the council **writes the statements, plans, trains, watches costs, measures, picks projects, builds teams, rewards.**

A6. Barriers to TQM implementation 🔴

#	Barrier (your slides)	One-line explanation
1	Lack of management commitment	Top management doesn't lead or resource the effort
2	Inability to change organisational culture	People resist new ways of working
3	Improper planning	No clear strategy, goals or timeline
4	Lack of continuous training and education	Employees lack the skills for TQM tools
5	Incompatible organisational structure; isolated individuals and departments	Silos block cross-functional teamwork

#	Barrier (your slides)	One-line explanation
6	Ineffective measurement techniques; poor access to data	Progress can't be tracked or decisions based on facts
7	Inadequate attention to internal and external customers	Customer needs are not the driver
8	Inadequate use of empowerment and teamwork	Employees are not involved in decisions
9	Failure to improve continuously	Improvement stops after early gains

Memory hook: **Commitment, Culture, Planning, Training, Structure, Measurement, Customer, Empowerment, Continuous improvement.**

Overcoming barriers (my addition for the "discuss" part of the answer): visible top-management commitment, a quality council, training for all levels, cross-functional teams, data-based measurement, and recognition of improvement.

SECTION B – 🟡 MODERATELY IMPORTANT

B1. Quality planning 🟡

- **Definition:** identifying which quality standards are relevant and deciding how to satisfy them.
- **Inputs:** quality policy, scope statement, product description, standards and regulations.
- **Methods:** benefit/cost analysis, benchmarking, flowcharting, design of experiments.
- **Outputs:** quality management plan, operational definitions, checklists.
- **Three quality management processes:** planning → assurance → control.
- **Juran trilogy:** quality planning (understand customer needs, develop features), quality control (meet goals during operations, measure deviation and act), quality improvement (break through to new performance levels).

B2. Historical review of TQM 🟡

Period	Milestone
B.C. era / Middle Ages	Skilled craftsmen and guilds; pride in quality workmanship
Industrial Revolution	Mass production; workers lost sense of ownership of quality
1924	Shewhart (Bell Labs) develops the control chart, beginning of SQC/SPC
1920s	Dodge and Romig develop acceptance sampling instead of 100% inspection
1946	American Society for Quality Control formed
1950	Deming teaches SQC/SPC to Japanese engineers and CEOs
1954	Juran teaches Japanese management their responsibility for quality
1960	First quality control circles in Japan
1970s	US managers learn from Japan's quality success
1980s	TQM principles and methods become popular
1990s	ISO 9000 becomes the worldwide QMS standard

B3. Principles of TQM 🟡

- From your slides: "total" means all members of the organisation make consistent efforts; TQM is customer-oriented management built on customer satisfaction, customer needs and expectations, and customer delight.
- Your CDP concept map shows eight quality management principles: **customer focus, leadership, people involvement, process approach, systems approach, continual improvement, factual (data-based) decision making, and mutually beneficial supplier relationships.** Learn these eight as the safest answer.
- Benefits of quality (slides): higher customer satisfaction, reliable products, better efficiency, more productivity and profit, better workforce morale, less wastage and inspection costs, more market share, better quality of life.

B4. Quality statements 🟡

Statement	Meaning
Vision	Short declaration of what the organisation hopes to be tomorrow
Mission	Statement of purpose: who we are, who our customers are, what we do, how we do it
Quality policy	Guide for everyone on how products and services must be provided to customers

- The CEO writes the quality policy after studying workforce feedback; the quality council approves it.
- When setting the policy, top management considers the improvement needed, desired customer satisfaction, development of people, needs of other interested parties, resources needed to go beyond ISO 9001, and contributions of suppliers and partners.
- The policy should be appropriate to the organisation's purpose, commit to compliance and continual improvement, provide a framework for quality objectives, be communicated and understood, and be reviewed for continuing suitability.

B5. Strategic planning 🟡

Seven steps: customer needs → customer positioning → predict the future → gap analysis → closing the gap → alignment → implementation.

Good strategic quality goals are: focused, concrete, based on statistical evidence, backed by a plan with resources, time-bound, and challenging yet achievable.

B6. Three spheres of quality 🟡 (from the classroom quiz)

The three spheres are **Quality Control, Quality Assurance and Quality Management**; management sits above and ties control and assurance together.

Sphere	Meaning	Example activities
Quality Control (QC)	Operational techniques to fulfil requirements; finds defects so they can be corrected; looks at the product	Monitoring process capability, control charts, acceptance sampling, reducing variability

Sphere	Meaning	Example activities
Quality Assurance (QA)	Process-driven activities guaranteeing quality; prevents defects by fixing the process	FMEA, concurrent engineering, experimental design, process improvement, reliability testing
Quality Management (QM)	Overarching management processes	Planning for improvement, quality culture, leadership, training, recognition, communication

Key difference to write: QC is detection (checks the finished product), QA is prevention (engineers the process), QM is the umbrella that plans and leads both.

SECTION C – 🟢 JUST READ

C1. Definition of quality 🟢

- Simple definition: meeting customer requirements; the ability to keep meeting them is reliability.
- Formula in slides: Q = P ÷ E (performance ÷ customer expectations).
- By source: ASQC/ANSI: totality of features that bear on the ability to satisfy needs; ISO 9000:2000: degree to which inherent characteristics fulfil requirements; Deming: predictable uniformity and dependability at low cost; Juran: fitness for use; Crosby: conformance to requirements.
- Five approaches: transcendent, product-based, user-based, production-based, value-based (value = benefits ÷ price).
- Consumer's view = quality of design and price; producer's view = quality of conformance and cost; the consumer's view dominates.

Extra gurus (skim only): Juran – fitness for use, Pareto principle, quality trilogy; Crosby – Quality is Free, Zero Defects, four absolutes (conformance to requirements, prevent errors, do it right the first time, measure by price of non-conformance).

Likely question skeletons (estimates, not guaranteed)

Likely question	Marks	Fast answer plan
Deming's philosophy + any five of 14 points	8–10	A1 template
Quality dimensions and four quality costs; effect on performance	8	A2 tables + A3 categories + performance argument
Leadership, quality council, strategic planning and barriers to TQM	7–8	A4, A5, B5, A6 headings, 2 lines each
Three spheres of quality with examples	4–6	B6 table
Quality statements (vision, mission, policy)	3–5	B4 table
Small quality-cost calculation	2–4	A3 formula and index

30-minute last-minute plan

1. 10 min – Deming: chain reaction, 4 profound-knowledge parts, five chosen points
2. 5 min – Quality cost table, total COQ formula and the performance argument
3. 5 min – Dimensions: P F R C D S R A R and the 7 service dimensions
4. 5 min – Council duties and the 9 barriers (use the memory hooks)
5. 5 min – Skim Section B headings and the three spheres table; skip Section C

Notes on source and limits

- Content is taken from the text of your Unit 1 PDF; items marked "my addition" or "my own example" are not from your slides.
- I read the slide text only, so diagrams (for example the PDSA cycle and the three-spheres figure) were not inspected visually. Check them in your slides.
- Importance levels follow your priority table; the mark estimates are mine.